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What if AI made the world's economic growth explode?

Markets for goods, services and financial assets, as well as labour, would be upended

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UNTIL 1700 the world economy did not really grow—it just stagnated. Over the previous 17 centuries global

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Growth just stagnated. Over the previous 17 centuries global output had expanded by 0.1% a year on average, a rate at which it takes nearly a millennium for production to double. Then spinning jennies started whirring and steam engines began to puff. Global growth quintupled to 0.5% a year between 1700 and 1820. By the end of the 19th century it had reached 1.9%. In the 20th century it averaged 2.8%, a rate at which production doubles every 25 years. Growth has not just become the norm; it has accelerated.

If the evangelists of Silicon Valley are to be believed, this bang is about to get bigger. They maintain that artificial general intelligence (AGI), capable of outperforming most people at most desk jobs, will soon lift annual GDP growth to 20-30% a year, or more. That may sound preposterous, but for most of human history, they point out, so was the idea that the economy would grow at all.

The likelihood that AI may soon make lots of workers redundant is well known. What is much less discussed is the hope that AI can set the world on a path of explosive growth. That would have profound consequences. Markets not just for labour, but also for goods, services and financial assets would be upended. Economists have been trying to think through how AGI could reshape the world. The picture that is emerging is perhaps counterintuitive and certainly mind-boggling.

It's the ideas, stupid!

Economies originally grew largely through the accumulation of people. Bigger harvests allowed more mouths to be fed; more farmers allowed for bigger harvests. But this form of growth did not raise living standards. Worse, famine was a constant menace. Thomas Malthus, an 18th-century economist, reasoned that population growth would inevitably outstrip agricultural yields, causing poverty. In fact, the reverse occurred: more people did not just eat more, but had more ideas, as well. Those ideas led both to higher output and, eventually, to lower fertility, which set output per person climbing. AGI, the theory runs, would allow for runaway innovation without any increase in population, supercharging growth in GDP per person.

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Most economists agree that AI has the potential to raise productivity and thus boost GDP growth. The burning question is, how much? Some predict only a marginal change. Daron Acemoglu of the Massachusetts Institute of Technology, for instance, estimates that AI will lift global GDP by no more than 1-2% in total over a decade. But this conclusion hinges on an assumption that only about 5% of tasks can be performed more cheaply by AI than by workers. That assumption, in turn, rests in part on research conducted in 2023, when AI was less capable.

More radical projections of AI's economic impact assume that much more of the world's economic output will eventually be automated as the technology improves and AGI is attained. Automating production then requires only sufficient energy and infrastructure—things that more investment can produce. Usually, investment-led growth is thought to hit diminishing returns. If you add machines but not workers, capital lies idle. But if machines get sufficiently good at replacing people, the only constraint on the accumulation of capital is capital itself. And adding AI power is much faster than waiting for the population to expand, argues Anson Ho of Epoch AI, a think-tank.

Even the total automation of production would not bring a growth explosion, however, according to a review of models by Philip Trammell, then of Oxford University, and Anton Korinek of the University of Virginia. Suppose production was fully automated, but technology did not improve. The economy would settle into a constant rate of growth, determined by the fraction of output that was saved and reinvested in building new machines.

Truly explosive growth requires AI to substitute for labour in the hardest task of all: making technology better. Will it be AI that delivers breakthroughs in biotechnology, green energy—and AI itself? AGI agents will, it is hoped, be able to execute complex, long-running tasks while interacting with computer interfaces. They will not just answer questions, but run projects. The AI Futures Project, a research group, forecasts that by the end of 2027, almost fully automated AI labs will be conducting scientific research. Sam Altman, the boss of OpenAI, has predicted that AI systems will probably start producing "novel insights" next year.

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Economists who study "endogenous" growth theory, which attempts to model the progress of technology, have long posited that if ideas beget more ideas with sufficient velocity, growth should increase without limit. Capital does not just accumulate; it becomes more useful. Progress is multiplicative. Humans have never crossed this threshold. In fact, some economists have suggested that ideas have become harder, not easier, to find over time. Human researchers must, for instance, master ever more material to reach the frontier of knowledge.

AGI might loosen those constraints. In Epoch's model, big early returns from automation are ploughed back into hardware and software research. Annual GDP growth passes 20% once AI can automate about a third of tasks, and keeps rising. The model, says Mr Ho, is "definitely wrong"—but it is hard to tell why. Economists think it is too optimistic about incentives to invest in research, whose benefits spill over to the economy, creating a collective-action problem. AI companies tell Mr Ho he is underestimating the feedback loops that kick in when AGI can improve itself—a process that, it is hoped, will ultimately bring about a superintelligence vastly more capable than any human.

The paroxysmal science

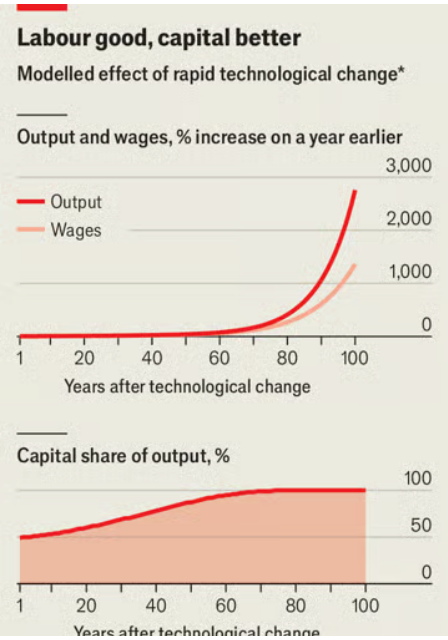
Assume those loops have maximum force and the economy becomes "information produced by information capital, which is produced by information, which in turn is producing information ever faster every year", as William Nordhaus, a Nobel laureate in economics, wrote in a paper in 2021. This brings about the "singularity"—a point when output becomes infinite. The singularity is really a counterargument: proof that the model must, eventually, be proved wrong. But even the first step on the journey, a big acceleration in growth, would be a profound event.

What would all this mean for workers? Humanity's first growth surge was not especially generous to them. An English

construction worker in 1800 earned the same real wages as one in 1230, according to Greg Clark of the University of Southern Denmark. The growing number of mouths to feed in effect nullified all the increase in output. Some historians argue that over the following 50 years or so, workers' living standards outright declined.

This time the worry is that workers become redundant. The price of running an AGI would place an upper bound on wages, since nobody would employ a worker if an AI could do the job for less. The bound would fall over time as technology improved. Assuming AI becomes sufficiently cheap and capable, people's only source of remuneration will be as rentiers—owners of capital. Mr Nordhaus and others have shown how, when labour and capital become sufficiently substitutable and capital accumulates, all income eventually accrues to the owners of capital. Hence the belief in Silicon Valley: you had better be rich when the explosion occurs.

A booming but workerless economy may be humanity's ultimate destination. But, argues Tyler Cowen of George Mason University, an economist who is largely bullish about AI, change will be slower than the underlying technology permits. "There's a lot of factors of production...the stronger the AI is, the more the weaknesses of the other factors bind you," he says. "It could be energy; it could be human stupidity; it could be regulation; it could be data constraints; it could just be institutional sluggishness." Another possibility is that even a superintelligence would run out of ideas. "AI may resolve a problem with the fishermen, but it wouldn't change what is in the pond," wrote Philippe Aghion of LSE and others in a working paper in 2017.



Hemmed in by such constraints, the economic impacts of AGI might not be quite as dramatic as the models suggest. As long as humans maintained an edge in some respects, people would toil alongside machines. And some of them would be extraordinarily

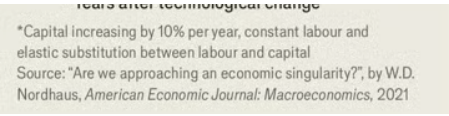


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well paid. In Mr Nordhaus's paper, less-than-perfect substitutability between labour and capital during an AI breakout leads to an explosion in wages. Strangely, wages still shrink as a share of the economy, since the economy is growing even faster (see chart). There is some evidence of this dynamic already within tech firms, which tend to pay superstar wages to top workers, even though the share of such firms' income that goes to owners is unusually high.

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Averages conceal variation. Explosive wages for superstars would not console those with more mundane desk-jobs, who would have to fall back on the parts of the economy that had not been animated. Suppose, despite AGI, that technological progress in robotics were halting. There would then be plenty of physical work requiring humans, from plumbing to coaching sports. These bits of the economy, like today's labour-intensive industries, would probably be affected by "Baumol's cost disease" (a wonderful affliction for workers) in which wages would grow despite a lack of gains in productivity.

In the classic case, named after an economist called William Baumol, wages grow to stop workers switching to industries in which productivity is surging. That would not apply with AGI, but other factors might produce Baumol-like effects. AI-owners and elite workers might spend a good deal of their new fortunes on labour-intensive services, for example. Think of today's wealthy, who shell out on lots of things that are hard to automate, from meals in restaurants to nannies. It is an optimistic vision: even those who are not superstars still benefit.

The non-rich would enjoy only selective abundance, however. Their purchasing power over anything that AI could produce

Their purchasing power over anything that AI could produce or improve would soar. Manufactured goods made in AI-run factories could be close to free; riveting digital entertainment might cost almost nothing; food prices, if AI worked out how to increase agricultural yields, could collapse. But the price of anything still labour-intensive—child care, say, or eating out—would need to rise in line with wages. Anyone who switched from today's knowledge work to a labour-intensive alternative might find that they could afford less of those bottle-necked goods and services than they can today.

Some worry that the Baumol effect would be so acute as to limit economic growth. When the price of something collapses, people buy more of it. But its share of consumer spending can still fall. Take food. In 1909 Americans bought 3,400 calories-worth of food per day (including waste), which cost 43% of their incomes. Today they buy 3,900 calories-worth, but that costs just 11% of their incomes. If prices fall faster than quantity increases, the measured economy becomes dominated by whatever it is that cannot be made more efficiently. "Growth may be constrained not by what we are good at but rather by what is essential and yet hard to improve," wrote Mr Aghion and his colleagues.

It is, however, important to keep Baumol effects in perspective, argues Dominic Coey of Meta. Even if they limit the economy's measured size, AGI could still bring vast change. Again, there is an echo of tech revolutions past. Smartphones and endless free online services have changed the world, but did not seem to affect growth much. And eventually, a superintelligence might solve bottlenecks too, for example by discovering new technologies that unlock greater energy supply, or accelerating progress in robotics.

What should you do if you think an explosion in economic growth is coming? The advice that leaps out from the models is simple: own capital, the returns to which are going to skyrocket. (It is not hard in Silicon Valley to find well-paid engineers glumly stashing away cash in preparation for a day when their labour is no longer valuable.) It is tricky, though, to know which assets to own. The reason is simple: extraordinarily high growth should mean extraordinarily high real interest rates.

Consider the financial forces that would kick in the moment an explosion in growth is on the cards. Massive investment would be wanted in data centres and energy production. You might think the amounts being invested today, such as OpenAI's $500bn "Stargate" project, are already extraordinary. But according to Epoch AI's model, the optimal investment in AI this year is 50 times more: $25trn. And that is just one part of the picture. A bigger economy would bring more demand

of the picture. A bigger economy would bring more demand for non-tech capital, too, to invest in things like infrastructure and bigger factories, as businesses expand to service higher demand. A race to invest would be on.

At the same time, the desire to save would be falling. On average, incomes would be about to rocket upwards. Economists tend to assume that people try to smooth their consumption over time: all else equal, they prefer to spend $100 today and $100 tomorrow than, say, $200 today and nothing tomorrow. Hence the need for savings, which can be invested to fuel growth. But a rocketing economy makes parsimony seem unnecessary. Lavish riches are coming, so why save? For that reason, noted Frank Ramsey, an early-20th-century economist, as growth rises, so do real interest rates, to entice carefree consumers to save some of the money they would otherwise be inclined to spend.



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For asset prices, this would mean a tug-of-war, Trevor Chow and colleagues argue in a recent working paper. Consider stocks. On the one hand, much higher interest rates would send the discount rate investors use to value future earnings soaring, and so slash the value of future cashflows. On the other hand, much faster growth, as long as a company was not itself at risk from AI, should lead to much higher future earnings. "The net effect on average stock prices is ambiguous," they conclude.

The strength of the Ramsey rule would be all-important: the greater the urge to even out consumption over time, the higher rates would rise if breakneck future growth is all but guaranteed. Unfortunately, there is no consensus on how strong the impulse to smooth spending really is. Macroeconomists tend to think it is so ingrained that rates typically rise faster than growth, causing stockmarkets to fall. Finance professors tend to believe the opposite: that growth outpaces rates.

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If that sounds too casino-like, there is an argument for simply depositing cash at the bank: an investor would then be able to take advantage of the higher interest rates without worrying about capital values. But if central banks failed to realise what was going on, and set interest rates lower than circumstances demanded, inflation would take off, eroding the value of cash. Land is another option. Its supply is fixed—and one theory is that a superintelligence might wish to carpet Earth with solar panels and data centres, bidding up land prices. Then again, land is among the most interest-rate sensitive assets. Imagine refinancing a mortgage at 30%.

More feathers or more hissing?

Higher interest rates would also complicate the picture for the world's indebted governments. Fast growth would ease their fiscal problems, but higher interest rates would make them worse. They might have to hand over lots of cash to wealthy bondholders, at a moment when job losses fuel demands for redistribution in the other direction—such as the universal handouts many in Silicon Valley expect to be necessary. Mr Cowen advocates a cheerful focus on the growing size of the pie, rather than worrying about how it is chopped up. But any country that is unable or unwilling to unleash AI-fuelled growth, while depending on global investors for capital, would face a brutal squeeze.

If investors thought all this was likely, asset prices would already be shifting accordingly. Yet, despite the sky-high valuations of tech firms, markets are very far from pricing in explosive growth. "Markets are not forecasting it with high probability," says Basil Halperin of Stanford, one of Mr Chow's co-authors. A draft paper released on July 15th by Isaiah Andrews and Maryam Farboodi of MIT finds that bond yields have on average declined around the release of new AI models by the likes of OpenAI and DeepSeek, rather than rising.

Silicon Valley, in other words, has yet to convince the world of its thesis. But the progress of AI has for the best part of a decade outpaced forecasts of when it would pass various benchmarks. You do not have to go back to 1700 to find someone you could surprise with humanity's subsequent progress: just imagine showing DeepSeek to a person from

2015. If the consensus about AI's effects on the economy is as behind-the-curve as most predictions of AI's capabilities have been, then investors—and everyone else—are in for a big surprise. The consequences of economic growth for human welfare, economist Robert Lucas once said, are so profound that "Once one starts to think about them, it is hard to think about anything else." As in so many other realms, the prospect of AGI compounds the phenomenon. ■

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